

Submitted To:

COURSE CODE: CAP538	COURSE INSTRUCTOR: GIRISH KUMAR
COURSE TITLE: Algorithms And Design Anlysis	ACADEMIC TASK NO. 1

Submitted By:

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ROLL NO. 05	GROUP: 1
SECTION: D2521	TERM: 2

Problem No. Even- Implement Binnary Search

```
#include <iostream>
using namespace std;
int binarySearch(int arr[], int low, int high, int key)
{
        if(low > high)
{
        return -1;
}
        int mid = low + (high - low) / 2;
        if(arr[mid] == key)
{
        return mid;
}
        else if (arr[mid] < key)
```

```
{
        return binarySearch(arr, mid + 1, high, key);
}
        else
{
        return binarySearch(arr, low, mid - 1, key);
}
}
int main() {
        int arr[] = {10, 20, 30, 40, 50};
        int size = sizeof(arr) / sizeof(arr[0]);
        int key;
        cout << "Enter element to search: ";
        cin >> key;
        int result = binarySearch(arr, 0, size - 1, key);
        if(result == -1){
                cout << "Element not found";
}
        else
        cout<< "element found at index " << result;
return 0;
}
```

binnsearch.cpp

File Edit View

```
#include <iostream>
using namespace std;
int binarySearch(int arr[], int low, int high, int key)
{
        if(low > high)
{
        return -1;
}
        int mid = low + (high - low) / 2;
        if(arr[mid] == key)
{
        return mid;
}
        else if (arr[mid] < key)
{
        return binarySearch(arr, mid + 1, high, key);
}
        else
{
        return binarySearch(arr, low, mid - 1, key);
}
}
int main() {
        int arr[] = {10, 20, 30, 40, 50};
        int size = sizeof(arr) / sizeof(arr[0]);
        int key;
        cout << "Enter element to search: ";
        cin >> key;
        int result = binarySearch(arr, 0, size - 1, key);
        if(result == -1){
                cout << "Element not found";
}
        else
        cout<< "element found at index " << result;
return 0;
}
```

Output:-

```
D:\THIRDSEMMCA\DAA\CAAOne>binnsearch
Enter element to search: 20
element found at index 1
D:\THIRDSEMMCA\DAA\CAAOne>
```